

Shubh Somani

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Summary — Machine learning (ML) engineer with hands-on experience building AI-powered applications spanning computer vision (CV), natural language processing (NLP), and retrieval-augmented generation (RAG). Skilled in deploying end-to-end deep learning systems and self-hosted infrastructure with a focus on measurable, real-world impact.

Education

Manipal University Jaipur

B.Tech Computer Science (Data Science)

GD Goenka Public School

CBSE 12th (PCM)

Certifications

- Microsoft Azure - From Zero to Hero
- Stanford - Unsupervised Learning, Recommenders, Reinforcement Learning
- Stanford - Supervised Machine Learning: Regression and Classification

Skills

Languages Python, Java, C, Batch/Shell
Frameworks PyTorch, TensorFlow, Scikit-Learn, FastAPI, LangChain
CV & ML Ultralytics (YOLOv8), MediaPipe, OpenCV, NumPy, Pandas, Matplotlib

Tools Docker, NGINX, Git, Azure Cloud, Proxmox VE, Tailscale VPN, Cloudflare Tunnels
Concepts Deep Learning, Computer Vision, NLP, RAG, REST APIs, DevOps

Experience

IEEE CS (Computer Society)

Oct 2024 – Apr 2026

Core Working Team (Technical)

Team Project : Hiring Assistant

- Architected an AI-powered hiring platform automating resume screening and candidate ranking using natural language processing (NLP), embeddings, and cosine similarity, cutting manual recruiter shortlisting effort.
- Built a scalable REST application programming interface (API) using FastAPI and Llama-based models to extract skills, match resumes against job descriptions, and auto-generate interview questions through a modular microservices architecture.

Projects

FullRag — Production-Ready RAG System

Apr 2026

- Engineered a full-stack, production-ready retrieval-augmented generation (RAG) platform using FastAPI (Python 3.12) and React + TypeScript, backed by PostgreSQL + pgvector for semantic similarity search across ingested documents.
- Implemented a modular 5-stage pipeline — document ingestion (PDF/text CLI), chunking, sentence-transformers embeddings, pgvector retrieval, and dual-provider LLM generation (Mistral Small / Groq llama-3.3) — reducing query-to-answer latency to under 1.3 seconds end-to-end.
- Shipped a continuous background evaluation loop, structured JSON observability/tracing, and a TTL-based response cache, improving system reliability and cutting redundant LLM calls.
- Containerized the full monorepo with Docker Compose (3 services: PostgreSQL + pgvector, FastAPI, nginx-served React SPA), with multi-stage Dockerfiles and Alembic migrations running on container startup.
- Authored a pytest test suite covering 5 modules (database, embeddings, chunker, API endpoints, evaluation) and built a Streamlit admin dashboard for real-time document browsing, query testing, and evaluation metrics inspection.

Plant Disease Classification with Noise Augmentation

Jan 2026

- Implemented and benchmarked 3 lightweight convolutional neural network (CNN) architectures, Mob-Res (3.51M params, 98.7% F1), MobileNetV2 (2.3M params, 99.3% F1), and ResNet-Path (1.2M params, 98.2% F1), evaluated on 2,662 held-out test images across a 56-class Plant Disease Expert dataset.
- Designed a noise augmentation pipeline applying Gaussian noise ($\sigma=0.01$), random horizontal flip, and 15° rotation over balanced per-class training splits, with cross-domain generalization validated on PlantVillage (CDVR).
- Engineered end-to-end training infrastructure across 4 Kaggle GPU notebooks using mixed-precision FP16 (AMP), Adam with $\beta_2=0.9$, early stopping (patience=20), and 128×128 normalized inputs on dual T4 GPUs.